



University of Niagara Falls – Canada

DAMO 640-08: Machine Learning (Winter Semester 2026)

Assignment 2: Analytical Model Choice & Segmentation

Group Member:

Bruno Sassi

Grace Dayrit

Nadera Ferguson

Tiago Ceolato

Dr. Touraj BaniRostam

March 1, 2026

TABLE OF CONTENTS

PART A – PROBLEM FRAMING AND PRE-ANALYSIS HYPOTHESES	2
PART B – EXPERIMENTAL DESIGN	5
PART C – CLASSIFICATION ANALYSIS	6
PART D – UNSUPERVISED LEARNING AND SEGMENTATION	7
AI Tool Use Disclosure	8
REFERENCES	9
APPENDIX	10

PART A – PROBLEM FRAMING AND PRE-ANALYSIS HYPOTHESES

This project uses the Customer_Behavior_Analytics dataset ($n = 600$) to (1) predict customer Channel (binary classification) from numerical spending/behavior/logistics features and (2) discover customer segments through clustering. Predicting Channel supports distribution planning and targeted marketing by identifying customers with systematically different buying patterns.

All predictors are numeric and include category spending (e.g., Fresh, Grocery, Milk), purchasing behavior (Order_Frequency, Avg_Order_Value), and logistics (Delivery_Distance). Because these drivers likely interact (e.g., *high order value + low frequency* vs. *moderate value + high frequency*), we expect nonlinear decision boundaries and threshold effects (e.g., “spend above X changes channel likelihood”), making model choice and preprocessing especially important.

1) Which model family will perform better (Ensemble vs. SVM) and why?

Hypothesis: 1- Ensemble will achieve higher mean ROC-AUC due to stronger nonlinear modeling of spending and behavior interactions. 2- SVM will show lower fold-to-fold variability (more stability) if scaling is handled correctly.

Tree ensembles naturally capture nonlinearities and interactions common in tabular customer data without manual feature engineering. They are robust to monotonic transformations and can model “if-then” structure (e.g., high Grocery *and* low Delivery_Distance) that often appears in channel behavior.

SVM may be more stable under cross-validation (lower variance across folds). SVM controls complexity through margin-based regularization (C) and, if using RBF, through kernel shape (γ). With proper feature scaling, SVM can generalize well and sometimes exhibits consistent performance across folds, especially if the true boundary is smooth. However, SVMs can be sensitive to scaling and hyperparameter choice. An overly flexible kernel may overfit, while an overly constrained setting may underfit.

Bias–variance rationale (what we expect to observe):

- Ensemble: tends to reduce variance (RF via averaging; GB via additive corrections) while still representing complex boundaries → likely higher average performance when interactions matter.

- SVM: regularization can yield stable CV results, but performance depends strongly on scaling and hyperparameters → may trade peak performance for stability.

2) Metric selection: Why ROC-AUC is most appropriate and why accuracy can mislead

Accuracy alone is not sufficient because it collapses performance into a single threshold and treats false positives and false negatives equally. In customer channel classification, the costs are rarely symmetric: misclassifying a high-value channel customer may cause lost revenue or poor allocation, while incorrectly labeling a customer as “key channel” can waste sales/ops effort.

Primary metric: ROC-AUC.

ROC-AUC evaluates how well the model ranks positives above negatives across all thresholds, making it appropriate when the decision threshold may change based on business constraints (e.g., capacity for targeted campaigns). ROC-AUC is also useful even if class balance is uncertain or moderate.

In addition to ROC-AUC, threshold-dependent metrics were reported to evaluate performance at the selected operating point. Recall measures the proportion of true channel members correctly identified, which is critical when missing them is costly. Precision reflects the efficiency of targeting, particularly when interventions are expensive. The F1-score provides a single balanced measure by combining precision and recall.

3) Clustering hypothesis: Expected segmentation structure (K-Means vs. DBSCAN)

Hypothesis: K-Means will yield clearer, business-friendly segments; DBSCAN will contribute by identifying outliers/noise and potentially small dense subgroups that K-Means absorbs.

- K-Means (distance/centroid-based) will likely produce interpretable segments driven by overall magnitude (spend and activity), which is typically useful for marketing personas. It assumes roughly spherical clusters and benefits from scaling (important because spending variables can dominate distances).
- DBSCAN (density-based) may better identify niche dense groups and label noise/outliers, which is valuable for detecting anomalies (e.g., very large customers or extreme logistics profiles). DBSCAN can struggle when cluster density varies and is sensitive to parameter choices (eps, min_samples).

Because features represent spending intensity, engagement, and logistics, we expect segmentation to reflect customer value and buying style. Expected segment types:

1. High spend + high frequency: loyal/high-value customers
2. Low spend + low frequency: low-engagement customers
3. Mixed behavior: e.g., high order value but low frequency (bulk buyers)
4. Outliers: unusual category concentration or extreme delivery distance

PART B – EXPERIMENTAL DESIGN

1. Data Splitting Strategy

The dataset ($n = 600$) contains a moderately balanced binary target ($0 = 330 / 1 = 270$). Although not severely imbalanced, preserving class proportions is important to avoid biased performance estimates. The data was split using a 70/30 stratified train–test split with a fixed random seed (`random_state = 42`) to ensure reproducibility. Stratification preserved class proportions across partitions. Training set (70%) was used exclusively for model selection and hyperparameter tuning and Test set (30%) was held out for final evaluation. This separation prevents data leakage and ensures unbiased estimation of generalization performance.

2. Cross-Validation Procedure

Model selection was performed using 5-fold stratified cross-validation on the training set only.

Within each fold 80% of the training data was used for model fitting and 20% was used for validation. Each observation served as validation exactly once and stratification ensured consistent class distribution across folds.

The primary selection metric was ROC-AUC, chosen for its threshold independence and suitability for ranking-based evaluation. For each hyperparameter configuration, both mean ROC-AUC and standard deviation across folds were recorded to evaluate performance and stability.

3. Model Selection and Hyperparameter Strategy

Model selection followed the assignment constraints by choosing one Ensemble model (Random Forest) and one SVM model (Linear SVM), tuning only a single hyperparameter per model family with a maximum of three candidate values. For Random Forest, the hyperparameter tuned was `max_depth` with values {5, 10, 20}, allowing control over model complexity and potential overfitting. For the Linear SVM, the regularization parameter `C` was tuned with values {0.1, 1, 10}, controlling the bias–variance trade-off. Because SVM is sensitive to feature scale, `StandardScaler` was applied prior to training to ensure fair coefficient estimation and stable optimization.

PCA Decision

PCA was not applied. The dataset contains only nine structured numerical features, and interpretability is valuable in a business setting. Since dimensionality is low and multicollinearity is not extreme, PCA was not necessary and would reduce interpretability without clear performance justification.

PART C – CLASSIFICATION ANALYSIS

Test Set Performance

Both models achieved strong and comparable discrimination (Table A1). The Random Forest (depth = 5) model's ROC-AUC was 0.884, while for the Linear SVM ($C = 0.1$) it was 0.890. The ROC curves (Figure A1) show very similar behavior, with SVM slightly dominating across most thresholds.

At the selected classification threshold, Random Forest achieved higher Recall (0.728), capturing more Channel 1 customers. Meanwhile, SVM achieved higher Precision (0.859), producing fewer false positives. F1-scores were nearly identical (~ 0.76), as can be seen in appendix's Table A2.

Contrary to the initial hypothesis that nonlinear structure might favor the Ensemble model, the results suggest that the class boundary is approximately linear. The best Random Forest configuration used shallow trees (depth = 5), indicating limited benefit from deeper nonlinear modeling.

The Linear SVM, with strong regularization ($C = 0.1$), slightly outperformed in ROC-AUC and showed more consistent performance across cross-validation folds. This suggests that the relationship between spending and channel membership is driven more by smooth magnitude differences than complex interaction effects.

Stability vs. performance trade-off

Although mean ROC-AUC was similar for both models, SVM exhibited lower cross-validation variability, indicating greater stability. When two models achieve comparable discrimination, lower variance and simpler decision boundaries generally favor deployment reliability. The SVM therefore demonstrates a better balance between performance and generalization consistency.

Business Error Considerations and Deployment Decision

The practical difference between models lies in error type. Random Forest produces fewer false negatives (higher Recall), capturing more target customers, while SVM produces fewer false positives (higher Precision), reducing inefficient allocation of resources. If the strategic priority is maximizing customer capture, Random Forest may be preferable. However, if operational efficiency and cost control are more critical, SVM is advantageous.

In conclusion, given the slightly higher ROC-AUC, greater stability, higher precision and comparable F1-score, the Linear SVM ($C = 0.1$) is recommended for deployment, as it offers strong discrimination with more consistent generalization and lower false-positive risk.

PART D – UNSUPERVISED LEARNING AND SEGMENTATION

K-Means Clustering

K-Means was evaluated for $k=2,3,4,5$ using silhouette score. The highest silhouette score occurred at $k = 2$ (Table A3), indicating that a two-cluster solution provides the best relative separation, although the score suggests moderate (not strong) cluster structure. Cluster comparison (Table A4) shows segmentation is primarily driven by spending magnitude and basket composition, not order frequency.

Table 1. Cluster characteristics

Cluster 0 – Higher-Value Segment	Cluster 1 – Lower-Spending Segment
Substantially higher spending across Fresh, Grocery, and Detergents_Paper	Lower overall spending
Much higher average order value (~2710 vs ~1398)	Significantly smaller basket size
Similar order frequency	Similar frequency but lower transaction value
Majority Channel ≈ 0.70	Majority Channel ≈ 0.31

This segment represents larger, higher-revenue customers with broader category coverage.	This group reflects smaller-scale or lower-intensity buyers.
--	--

Note. The clusters are interpretable and strategically actionable, separating high-value from lower-value customers.

DBSCAN Results

DBSCAN was evaluated using $\text{eps} = 0.5$ and $\text{eps} = 1.0$. In both cases zero clusters and 600 noise points were detected. DBSCAN classified all observations as noise, indicating no dense local groupings in feature space. This suggests that customer variation is continuous and diffuse, rather than organized into dense pockets. The dataset lacks strong density-based structure suitable for DBSCAN segmentation.

K-Means vs. DBSCAN

The two methods reveal different structural assumptions. While k-Means partitions data globally and successfully identified meaningful value-based segments, DBSCAN searches for high-density regions and failed to detect clusters, implying that customers vary gradually rather than forming compact dense groups. In this dataset, segmentation is better characterized by magnitude-based partitioning rather than density-based grouping. Therefore, k-Means is more appropriate for strategic customer segmentation.

Conclusion

The supervised analysis demonstrated strong predictive separation between channels (ROC-AUC ≈ 0.89), with the Linear SVM offering slightly better discrimination, greater stability, and higher precision, making it the recommended deployment model. Unsupervised analysis revealed that customer structure is primarily driven by spending intensity. K-Means identified two interpretable segments separating higher-value from lower-value customers, while DBSCAN failed to detect density-based clusters, indicating a continuous rather than locally concentrated distribution. Together, these results suggest that customer channel behavior is both predictably separable and strategically segmentable, providing actionable insights for targeted marketing and resource allocation.

AI Tool Use Disclosure

Generative AI (ChatGPT) was used to assist with improving clarity of written explanations, refining analytical interpretations, and structuring sections of the report (e.g., executive summary and

conclusion). All modeling decisions, code implementation, experimental design, and interpretation of results were conducted by the student. AI assistance was limited to language refinement and organizational support.

REFERENCES

- Hex.Tech. (n.d.). Comparing density-based methods. <https://hex.tech/blog/comparing-density-based-methods/#choosing-your-approach>
- IBM. (2020). What is a support vector machine? <https://www.ibm.com/think/topics/support-vector-machine>
- IBM. (n.d.-a). What is ensemble learning? IBM Think. <https://www.ibm.com/think/topics/ensemble-learning>
- NB Data. (n.d.). Comparing model ensembling: Bagging, boosting, and stacking. <https://www.nb-data.com/p/comparing-model-ensembling-bagging>
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Support Vector Machines. In Scikit-learn: Machine learning in Python. <https://scikit-learn.org/stable/modules/svm.html>
- Scikit-learn developers. (n.d.-a). Ensemble methods. <https://scikit-learn.org/stable/modules/ensemble.html>
- Scikit-learn developers. (n.d.-b). Unsupervised learning. https://scikit-learn.org/stable/unsupervised_learning.html

APPENDIX

Table A1. Confusion Matrices

Random Forest		SVM	
84	15	90	9
22	59	26	55

Table A2. Test Metrics – best performing models

Model	ROC-AUC	Precision	Recall	F1-score
Random Forest	0.884	0.797	0.728	0.761
SVM	0.89	0.859	0.679	0.759

Figure A1. ROC Curve Comparison

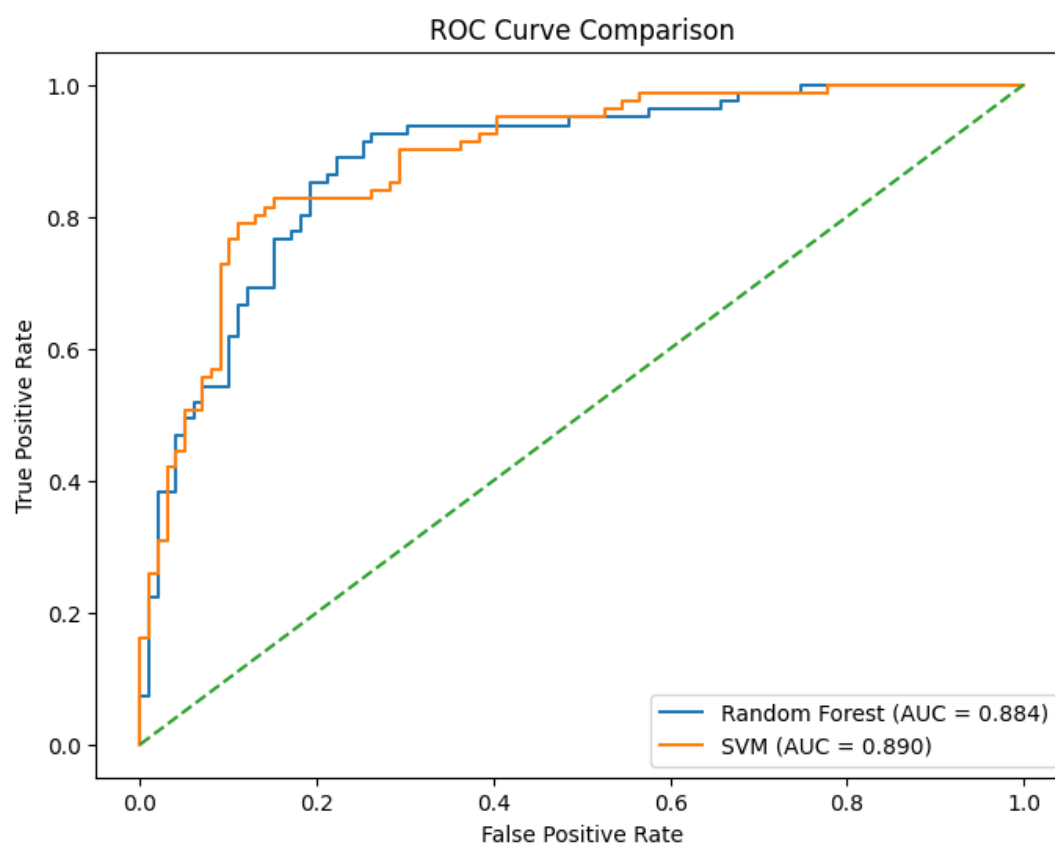


Table A3. Silhouette score for k-means clusters

k	score
2	0.172
3	0.145
4	0.13
5	0.112

Table A4. Clusters mean feature values

	Cluster 0	Cluster 1
Fresh	3839.991	2443.892
Milk	1928.896	1964.959
Grocery	4289.505	1903.369
Frozen	1637.741	1232.657
Detergents_Paper	1968.024	835.9742
Delicatessen	797.2028	950.4691
Order_Frequency	12.19811	11.87964
Avg_Order_Value	2710.255	1397.825
Delivery_Distance	17.47736	18.22758
Channel	0.698113	0.314433

Figure A2. Visualization of k-means clusters (with PCA)